

Skills Summary

Circles and Parabolas in Standard Form

Use this reference while completing your worksheet or when studying.

Skill 1 — Completing the square on a circle

Standard (centre-radius) form: $(x - h)^2 + (y - k)^2 = r^2$, with centre (h, k) and radius r . The number on the right is r^2 , never r .

The move: $x^2 + bx$ becomes a perfect square when you add $(\frac{b}{2})^2$, and it then equals $(x + \frac{b}{2})^2$.

The method: group the x terms and the y terms, move the constant to the right, complete both squares, and add each new number to the right-hand side as well. If the squared terms carry a coefficient, divide the whole equation by it first.

Example: Write $x^2 + y^2 + 2x - 8y + 8 = 0$ in standard form.

Step 1. The equation has both squares with coefficient 1 and no way to read a centre off it, so group each variable and complete the square twice.

Step 2. $(x^2 + 2x) + (y^2 - 8y) = -8$; half of 2 is 1 so add 1, half of -8 is -4 so add 16, and both go on the right: $-8 + 1 + 16 = 9$.

$$\Rightarrow (x + 1)^2 + (y - 4)^2 = 9, \quad \text{centre } (-1, 4), \text{ radius } 3.$$

Try it: Write $x^2 + y^2 - 4x - 6y - 3 = 0$ in standard form.

check: $(x - 2)^2 + (y - 3)^2 = 16$, centre $(2, 3)$, radius 4

Skill 2 — Completing the square on a quadratic

Vertex form: $y = a(x - h)^2 + k$, with vertex (h, k) ; the parabola opens up when $a > 0$ and down when $a < 0$.

The balancing is different from a circle. There is only one expression here, not two sides, so the number you add must be subtracted again straight away: $x^2 + bx + c$ becomes $(x + \frac{b}{2})^2 - (\frac{b}{2})^2 + c$.

If a coefficient sits in front of x^2 , factor it out of the x terms *only* — then whatever you add inside the bracket is multiplied by that coefficient before it is subtracted back off. Completing the square in y instead of x works identically and gives a parabola lying on its side.

Example: Write $y = x^2 + 10x + 18$ in vertex form.

Step 1. Nothing about the vertex is visible in this form, and the coefficient of x^2 is 1, so complete the square directly on the x terms.

Step 2. Half of 10 is 5 and $5^2 = 25$, so add 25 and take it away again: $y = (x^2 + 10x + 25) - 25 + 18$.

$$\Rightarrow y = (x + 5)^2 - 7, \quad \text{vertex } (-5, -7).$$

Try it: Write $y = x^2 - 6x + 2$ in vertex form.

check: $(x-3)^2 - 7 = x^2 - 6x + 2$

Skill 3 — Reading the graph off the finished form

Once a circle is in standard form you can read the centre and radius straight off, and that is what makes every other question answerable.

Signs: $(x + 4)^2$ is $(x - (-4))^2$, so $h = -4$. Flip the printed sign.

Is a point on the circle? Measure its distance d from the centre,

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

and compare it with r : less means inside, equal means on the circle, greater means outside.

Example: For $(x - 4)^2 + (y + 2)^2 = 36$, give the radius and decide where the point $(4, 4)$ lies.

Step 1. The form gives the centre $(4, -2)$ and $r^2 = 36$ for free, so the question becomes a distance comparison rather than anything algebraic.

Step 2. Radius: $\sqrt{36}$. Distance from $(4, -2)$ to $(4, 4)$: the two points share an x -coordinate, so it is just $4 - (-2)$.

$\Rightarrow r = 6$ and the distance is 6 , so the point sits exactly *on* the circle.

Try it: For $(x + 3)^2 + (y - 1)^2 = 25$, how far is $(1, 4)$ from the centre, and where does it lie?

check: distance 5, equal to the radius, so it is on the circle